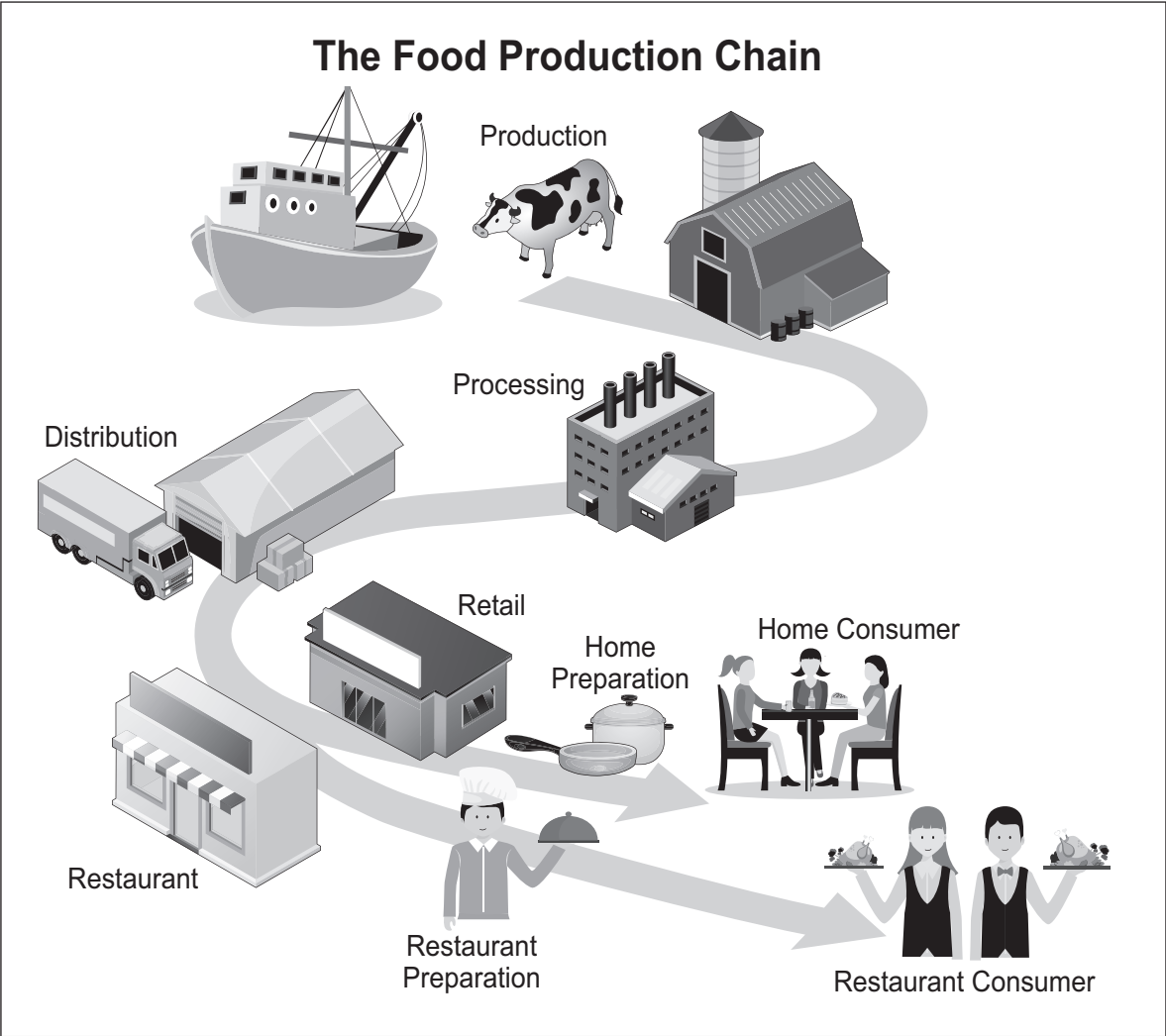


2. The food industry uses fruits, vegetables, grains and raw meat and processes these into the food products that are sold in supermarkets. Food production often involves cleaning and cooking raw foods, as well as the addition of additives and other ingredients.

One purpose of food production processes is to create products with longer shelf-lives than raw food ingredients.



<https://encrypted-tbn2.gstatic.com/images?q=tbn:ANd9GcRVLh-HrTveuT6xs04UgN3GbUEnX8XwCN8JHsQv4zLXGvMZfs3aFA>

The food industry must take precautions to prevent consumers becoming ill from food poisoning.

- (a) Complete the table below which shows some of the methods of controlling bacteria in the food industry. [3]

Method	Effect on bacteria
refrigeration	slows down bacterial growth
freezing	
.....	kills bacteria
.....	dries bacteria to prevent growth

(b) Sometimes food becomes contaminated with bacteria, causing food poisoning.

(i) State **one** symptom of food poisoning.

[1]

.....

(ii) Name **one** type of bacteria that causes food poisoning.

[1]

.....

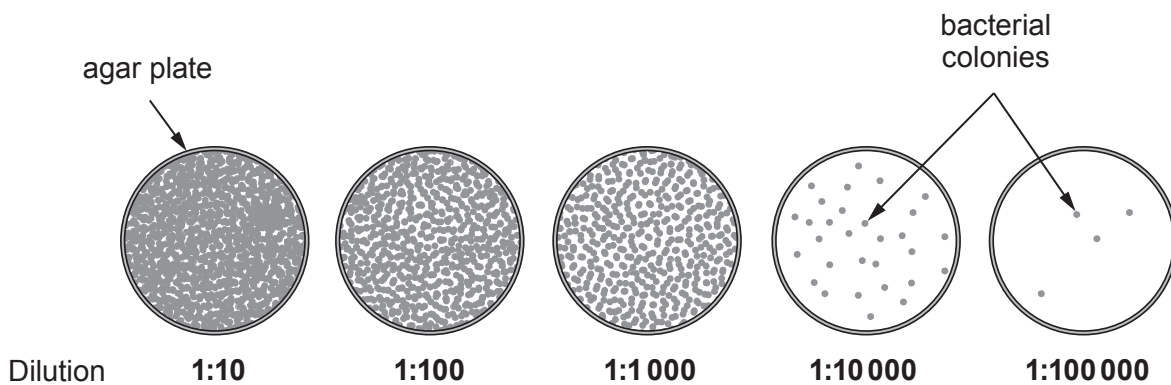
(iii) State the harmful substances produced by bacteria as they grow.

[1]

.....

(c) Whenever an outbreak of food poisoning occurs, environmental health officers test samples of food for the presence of bacteria.

(i) To be able to count the number of bacterial colonies a series of dilutions is carried out as shown in the diagram below.



The 1:10 000 dilution was used to calculate the number of bacteria present.

I Explain why using a dilution of 1:10 will not provide valid results.

[2]

.....

II Give **one** reason why the 1:100 000 dilution was also not used to calculate the number of bacteria present.

[1]

.....

Examiner
only

- (ii) In an investigation, six agar plates were prepared by adding 1 cm³ of solution from a 1:10 000 dilution. The agar plates were incubated for 72 hours. The results are shown below.

Plate number	1	2	3	4	5	6
Number of colonies	35	35	25	15	30	25

I State which plate has produced an anomalous result. [1]
.....

II Use the equation:

$$\text{number of bacteria/cm}^3 = \text{mean number of colonies} \times 10\,000$$

to calculate the number of bacteria/cm³ in the original sample. [3]

Number of bacteria/cm³ =

13